

■ Informations sur le document

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■ Résumé détaillé

Integrated Project Building Multi-Agent AI Systems Prof. At least one agent must use a deep learning model you trained as a functional tool — not decoration. The system must include error handling, logging, and a human-in-the-loop checkpoint. Technical Stack Agent framework CrewAI (primary) LangGraph for stretch goal Gemini free API / Ollama No personal spending required Consistent with course Agent Foundations & Working single-agent prototype, architecture doc, domain + dataset chosen DL Model as Agent Tool Trained model with evaluation, model wrapped as tool, agent uses model in its Multi-Agent Orchestration 2+ agents + orchestrator, human-in-the-loop, end-to-end demo Evaluation & Defense Guardrails, edge-case testing, final report, live demo + 10-min Q&A Suggested Domains Choose one (or propose your own with approval). You must understand and explain every line of your code during the oral defense. “The goal is not to build the most complex system. It is to build a system where every component exists for a reason, every agent has a job, and you can explain why.”

■ Points clés

- Integrated Project Building Multi-Agent AI Systems Prof.
- Hakim Hafidi | 4 weeks | Teams of 3 Design, build, and evaluate a multi-agent system where specialized AI agents collaborate to solve a real-world problem.

- At least one agent must use a deep learning model you trained as a functional tool — not decoration.
 - The system must include error handling, logging, and a human-in-the-loop checkpoint.
 - System Design & Architecture Agent roles justified, communication diagram, design rationale
DL Model Integration Appropriate model, rigorous evaluation, meaningful integration Working
Multi-Agent System Runs end-to-end, genuine collaboration, code quality, HITL Evaluation &
Robustness Edge cases tested, honest failure analysis, guardrails, logging Presentation &
Report Live demo works, can defend decisions, 8–12 page report Minimum Requirements •
Agents: 2 specialists + 1 orchestrator, each with distinct role and tools • DL model: PyTorch,
trained/fine-tuned by you, with evaluation (accuracy, confusion matrix) • Tools: At least 2
(including your DL model), with clear input/output schemas • Human-in-the-loop: At least 1
checkpoint requiring human approval • Error handling: Tool failures, invalid inputs, unexpected
LLM outputs — no crashes • Logging: Every agent action logged with timestamps (JSON) •
Reproducibility: README with setup instructions Submission: GitHub repo + trained model +
PDF report (8–12 pages) + demo video (3–5 min) + slides.
 - Academic integrity: LLM assistants encouraged as tools.
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